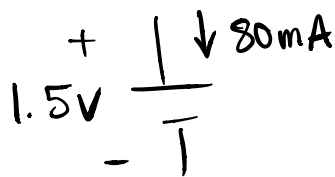


.. start w/ battery

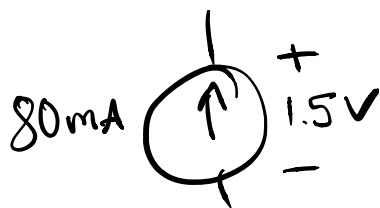


← we know this because they're in series

∴ $P = +VI$
 $= +1.5 \cdot 0.08 = \underline{0.12 \text{ W}}$

positive power → dissipating

.. current source



← because they're in parallel

$P = -VI$

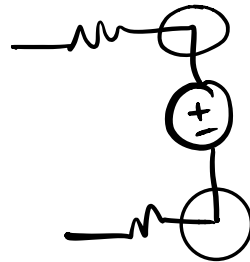
$= -1.5 \cdot 0.08$

$P = -0.12 \text{ W}$ OR $\underline{0.12 \text{ W generated}}$

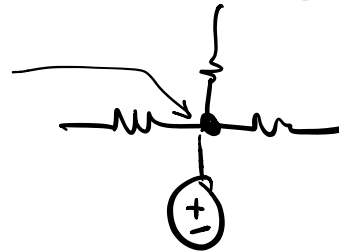
of course $P_{\text{diss}} = P_{\text{gen}}$

Nodes

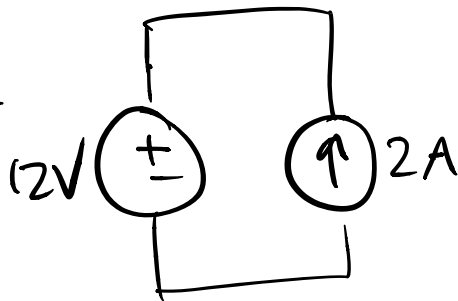
two kinds: elementary



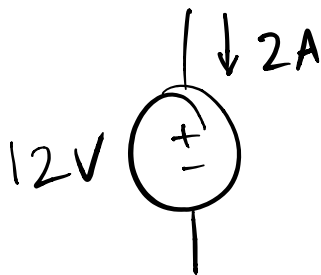
principal



ex:



.. find power in each element

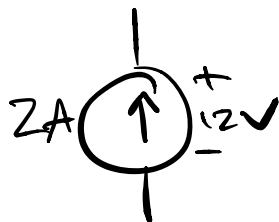


↙ voltage drop

$$P = +VI$$

$$= +12 \cdot 2 = +24W$$

(dissipated)



↘ voltage rise

$$P = -VI$$

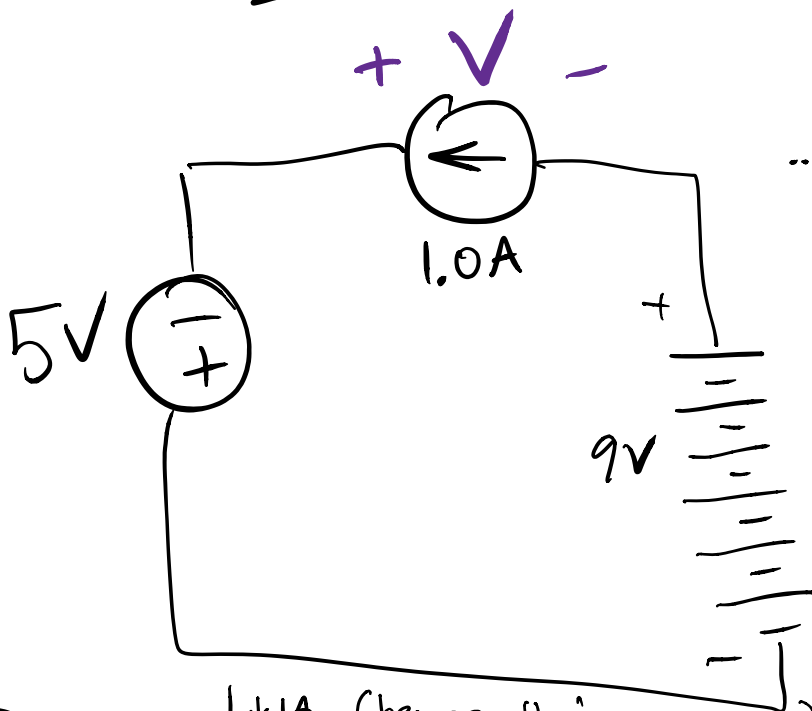
$$= -12 \cdot 2 = -24W$$

Conservation of Power

.. in a closed circuit, $\sum P = 0$

$$\text{OR } \sum P_{\text{diss.}} = \sum P_{\text{gen.}}$$

$$\sum P = +24 + -24 = 0 \quad \checkmark$$



.. determine V and all powers; verify that power is conserved

P_{5V} : $5V$ $\downarrow 1A$ (because they're in series)

$$P_{5V} : 5V \begin{matrix} \ominus \\ + \end{matrix}$$

$$P_{5V} = -V I = -5 \cdot 1 = \underline{\underline{-5W}}$$

$$P_{9V} : \begin{matrix} + \\ 9V \\ - \end{matrix} \uparrow 1A$$

$$P_{9V} = -V I = -9 \cdot 1 = \underline{\underline{-9W}}$$

P_{1A} : we don't yet know V !

.. we can determine P_{IA} by conservation of power

$$\sum P = P_{5V} + P_{9V} + P_{IA} = 0$$

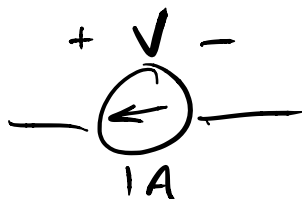
$$= -5 + -9 + P_{IA} = 0$$

$$\underline{P_{IA} = 14W}$$

Verify: $\sum P_{diss.} = P_{IA} = 14W$

$$\sum P_{gen.} = P_{5V} + P_{9V} = 14W$$

$$\sum P_{diss.} = \sum P_{gen.} \quad \checkmark$$



$$P_{IA} = \overset{\text{rise}}{\downarrow} -VI$$

$$V = \frac{-P}{I} = \frac{-14}{1}$$

$$\underline{V = -14V}$$

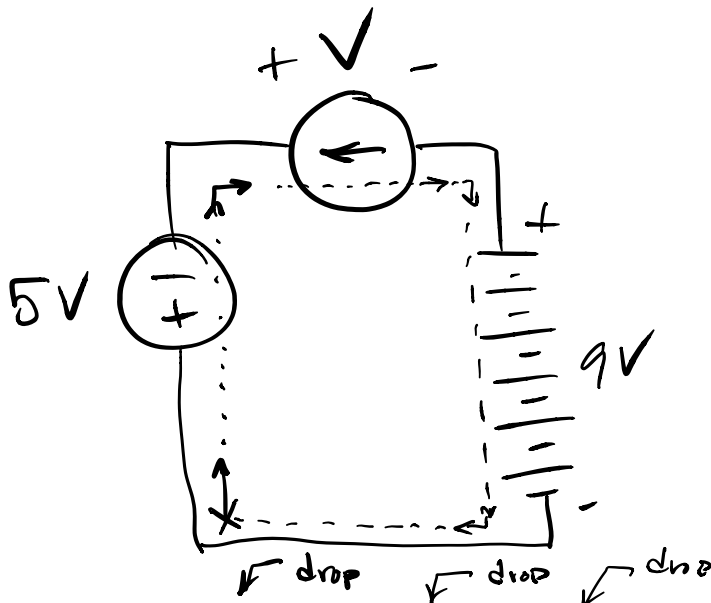
.. is there a better way? Yep.

Kirchoff's Voltage Law

.. sum of voltage drops around a closed loop equals Zero

.. the key: pick a starting point and direction and stick with it

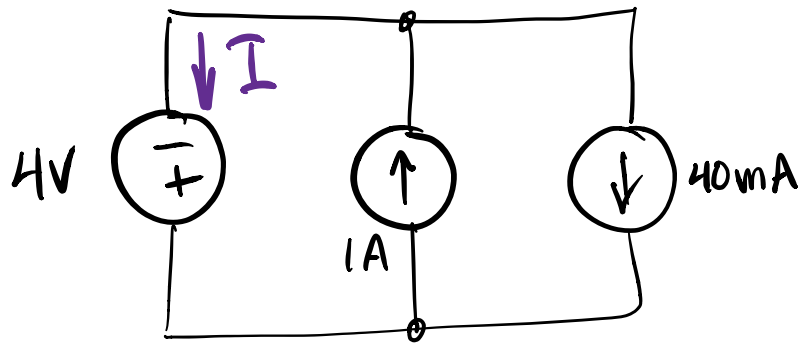
Joe Tritschler: CW
LLH
add drops



$$+5 + V + 9 = 0$$

$$V = -9 - 5 = \underline{-14V}$$

ex.)



.. determine I and all powers

1A: $\begin{array}{c} | \\ \uparrow \\ \text{1A} \end{array} \begin{array}{c} - \\ 4V \\ + \end{array}$ ← we know this voltage
because they're in parallel

$$P_{1A} = \overset{\downarrow \text{drop}}{+V} I = 4 \cdot 1 = \underline{4W}$$

40mA: $\begin{array}{c} | \\ \downarrow \\ \text{40mA} \end{array} \begin{array}{c} - \\ 4V \\ + \end{array}$ ← same deal!

$$P_{40mA} = \overset{\downarrow \text{rise}}{-V} I$$

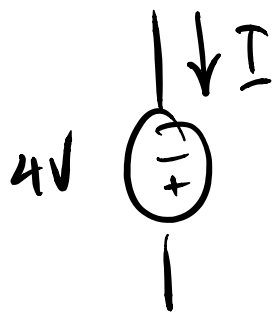
$$= -4 \cdot 0.04 = -0.160W$$

or -160mW
or 160mW gen.

Conservation of power:

$$\sum P = P_{4V} + P_{1A} + P_{40mA} = 0$$

$$P_{4V} = -P_{1A} - P_{40mA} = -4 - -0.16 = \underline{-3.84W}$$



$$P_{4V} = -V I$$

rise

$$I = \frac{P_{4V}}{-V} = \frac{-3.84}{-4}$$

$$\underline{I = 0.96 \text{ A}}$$

better way: Kirchoff's Current Law

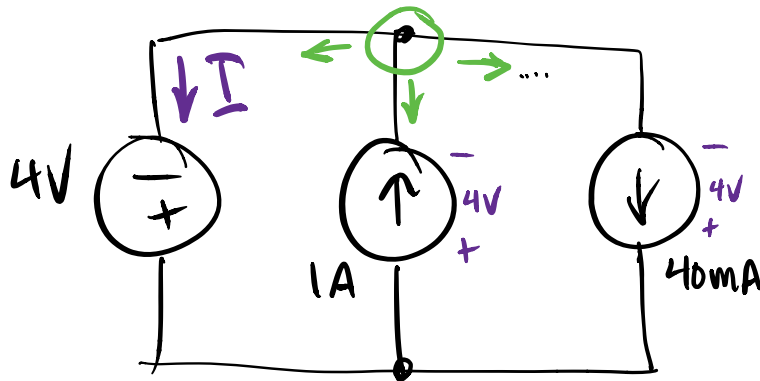
.. sum of currents leaving a node must sum to zero

.. alt: currents entering

Joe Tritschler : always leaving

the key: pick a ^{principal} node of interest

.. this example: two principal nodes,
both carry same currents, so doesn't
matter which you use!



.. pick top node

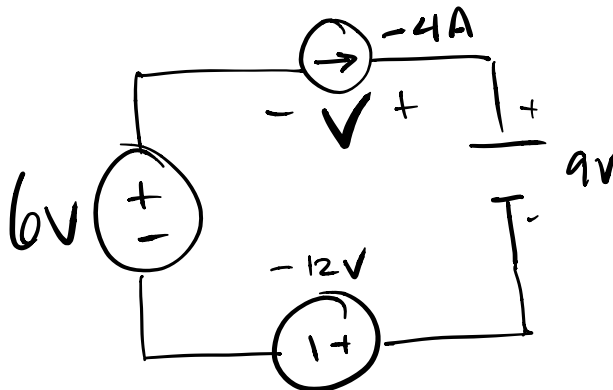
entering

$$\sum I = I + -1 + 0.04 = 0$$

$$I = 1 - 0.04 = \underline{0.96 \text{ A}}$$

$$P_{4V} = \overset{\text{rise}}{-VI} = -4 \cdot 0.96 = \underline{-3.84 \text{ W}}$$

homework: determine power in each element
and missing voltage ✓



.. use Conservation of power

.. then use KVL